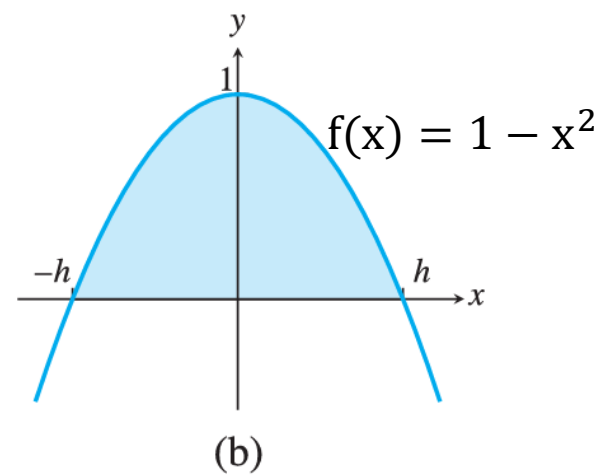
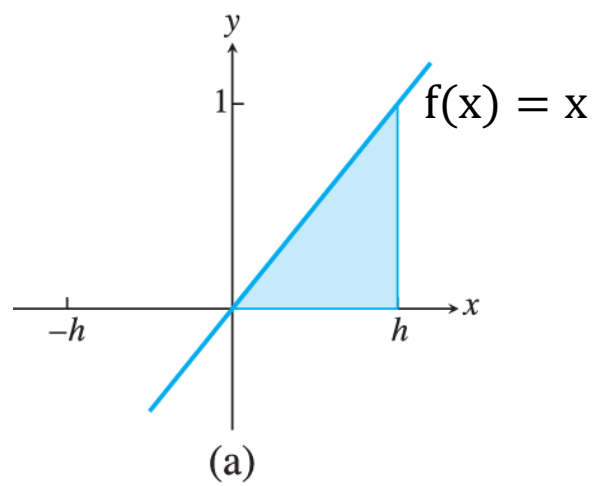
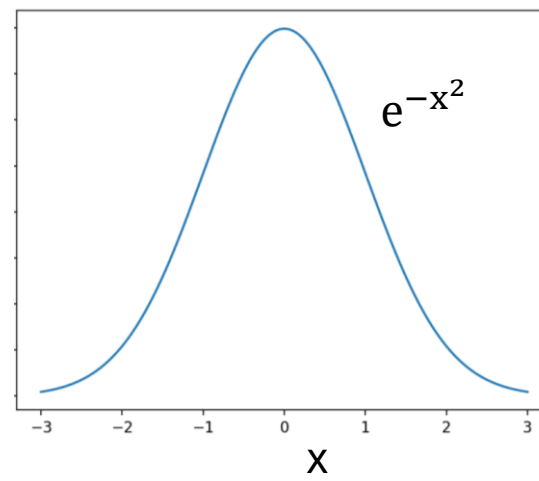


Integration

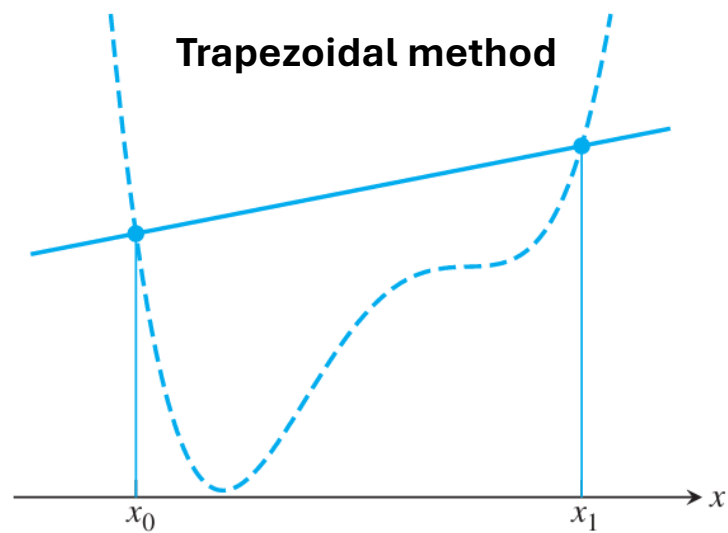
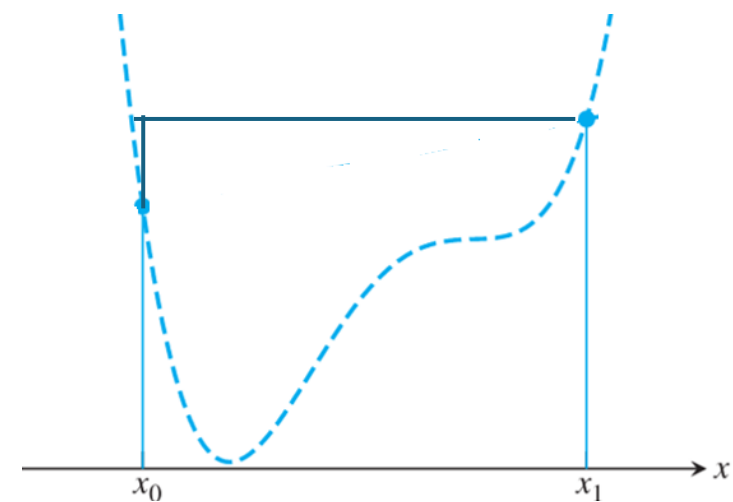
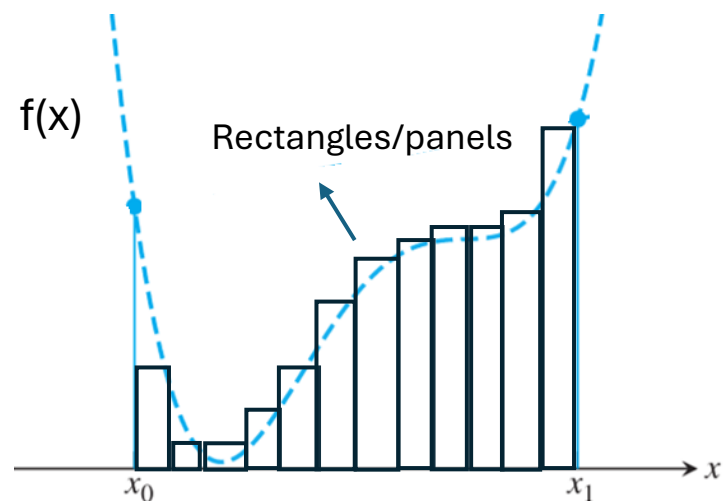


$$\int_0^t e^{-x^2} dx$$

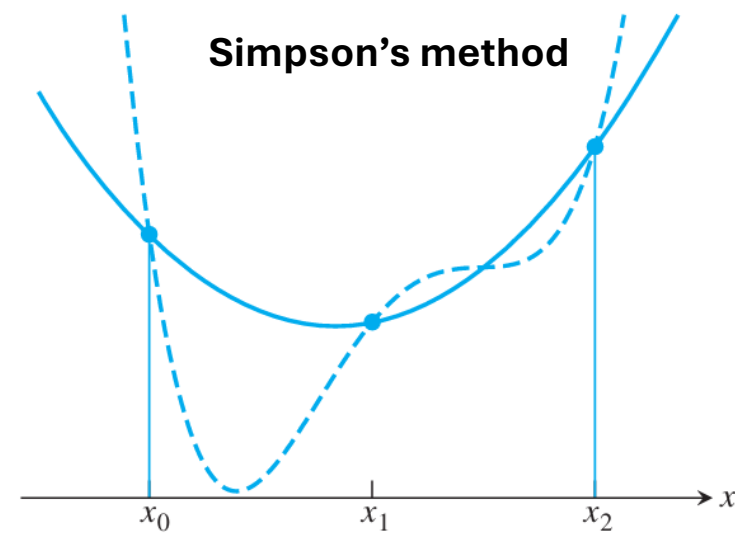
Gaussian function



Numerical Integration:

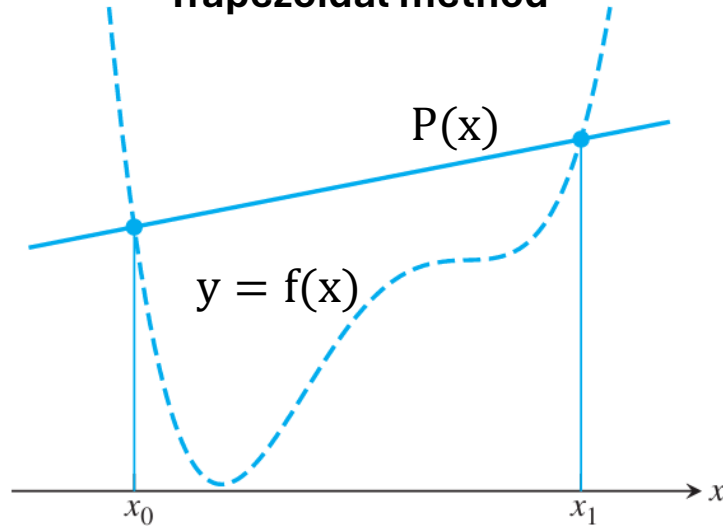


(a)



(b)

Trapezoidal method



$$f(x) = P(x) + E(x)$$

$P(x)$ = interpolating polynomial

$E(x)$ = interpolation error

$$P(x) = f(x_0) \frac{(x - x_1)}{(x_0 - x_1)} + f(x_1) \frac{(x - x_0)}{(x_1 - x_0)}$$

(Lagrange interpolation)

$$\int_{x_0}^{x_1} f(x) dx = \int_{x_0}^{x_1} P(x) dx + \int_{x_0}^{x_1} E(x) dx.$$

$$\begin{aligned} \int_{x_0}^{x_1} P(x) dx &= y_0 \int_{x_0}^{x_1} \frac{x - x_1}{x_0 - x_1} dx + y_1 \int_{x_0}^{x_1} \frac{x - x_0}{x_1 - x_0} dx \\ &= y_0 \frac{h}{2} + y_1 \frac{h}{2} = h \frac{y_0 + y_1}{2}, \quad \boxed{h = x_1 - x_0} \end{aligned}$$

substituting $w = -x + x_1$ into the first integral

$$\int_{x_0}^{x_1} \frac{x - x_1}{x_0 - x_1} dx = \int_h^0 \frac{-w}{-h} (-dw) = \int_0^h \frac{w}{h} dw = \frac{h}{2},$$

Trapezoid Rule

$$\int_{x_0}^{x_1} f(x) dx = \frac{h}{2} (y_0 + y_1)$$

Error in Trapezoidal method

$$\int_{x_0}^{x_1} f(x) dx = \int_{m-h/2}^{m+h/2} f(x) dx$$

Taylor remainder theorem:

$$f(x) = f(m) + (x - m)f'(m) + R(x) \text{ [Remainder]}$$

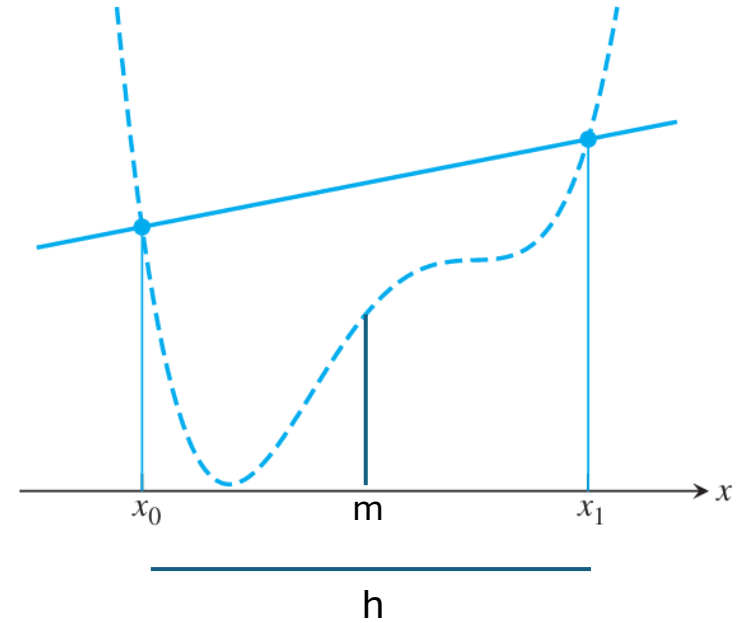
$$\text{where, } R(x) = \frac{(x - m)^2}{2} f''(c) \text{ and } c \text{ is a constant between } x \text{ and } m$$

Therefore, the exact integral

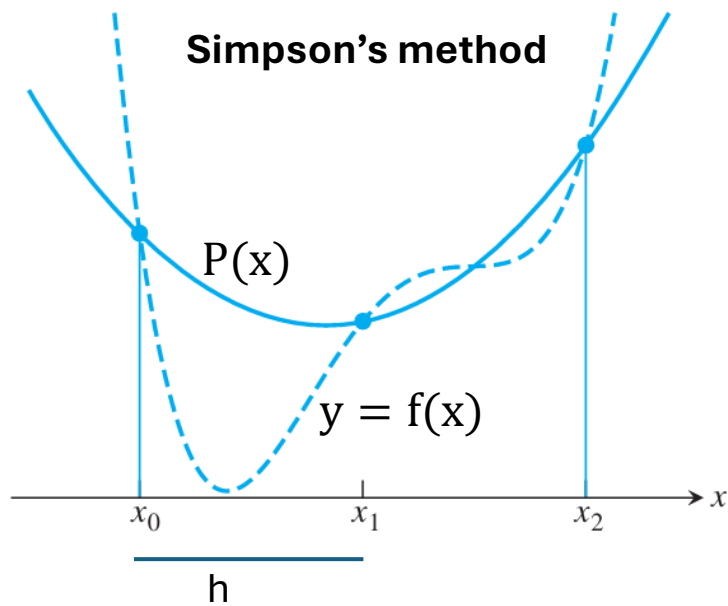
$$\begin{aligned} I &= \int_{m-h/2}^{m+h/2} f(x) dx = \int_{m-h/2}^{m+h/2} \left(f(m) + (x - m)f'(m) + \frac{(x-m)^2}{2} f''(c) \right) dx \\ &= hf(m) + \frac{h^3}{24} f''(c) \end{aligned}$$

Using the trapezoidal rule

$$\begin{aligned} S &= \int_{m-h/2}^{m+h/2} f(x) dx = \frac{h}{2} [f(m - h/2) + f(m + h/2)] \\ &= hf(m) + \frac{h^3}{8} f''(c) \text{ (using the Taylor remainder theorem)} \end{aligned}$$



$$\begin{aligned} \text{Error, } \int_{m-h/2}^{m+h/2} E(x) dx &= I - S \\ &= -\frac{h^3}{12} f''(c) \end{aligned}$$



$$f(x) = P(x) + E(x)$$

$P(x)$ = interpolating polynomial

$E(x)$ = interpolation error

$$P(x) = y_0 \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} + y_1 \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} + y_2 \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$\int_{x_0}^{x_1} f(x) dx = \int_{x_0}^{x_1} P(x) dx + \int_{x_0}^{x_1} E(x) dx.$$

$$\int_{x_0}^{x_2} P(x) dx = y_0 \int_{x_0}^{x_2} \frac{(x - x_1)(x - x_2) dx}{(x_0 - x_1)(x_0 - x_2)} + y_1 \int_{x_0}^{x_2} \frac{(x - x_0)(x - x_2) dx}{(x_1 - x_0)(x_1 - x_2)} + y_2 \int_{x_0}^{x_2} \frac{(x - x_0)(x - x_1) dx}{(x_2 - x_0)(x_2 - x_1)}$$

$$= y_0 \frac{h}{3} + y_1 \frac{4h}{3} + y_2 \frac{h}{3}.$$

$$h = x_2 - x_1 = x_1 - x_0$$

Simpson's Rule

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} (y_0 + 4y_1 + y_2)$$

Following the exact procedure for trapezoidal method, one can show that

$$\text{Error, } \int_{m-h/2}^{m+h/2} E(x) dx = -\frac{h^5}{90} f''''(c)$$

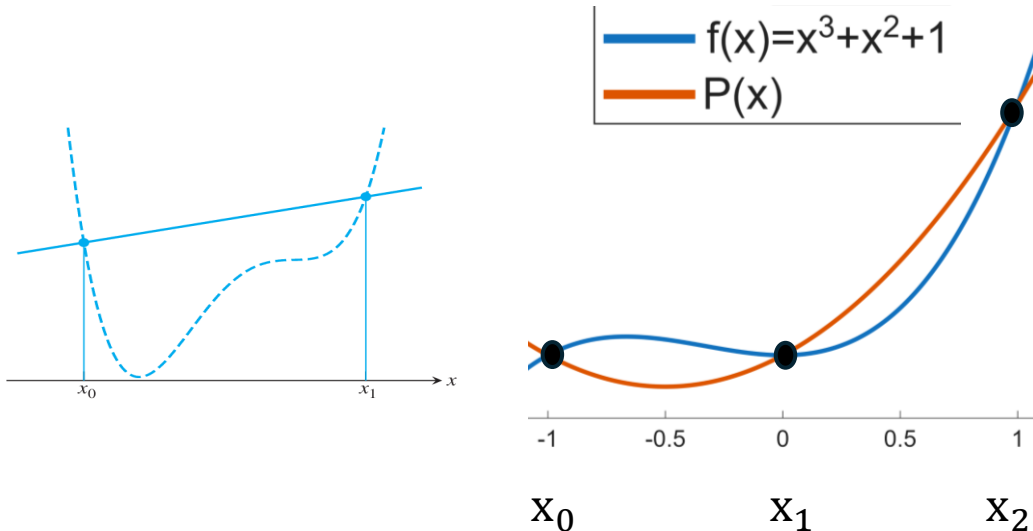
Degree of precision

The **degree of precision** of a numerical integration method is the greatest integer k for which all degree k or less polynomials are integrated exactly by the method. $\square$

The error in the trapezoidal method is $\frac{-h^3}{12} f''(c)$.

So, if $f(x)$ is polynomial of degree $n \leq 1$, the error is zero and the integral is exact.

Hence, the degree of precision of trapezoidal method is 1.



Similarly, the degree of precision of the Simpson's method is 3, since the error is $\frac{-h^5}{90} f''''(c)$.

Let $f(x) = ax^3 + bx^2 + cx + d$

Take an interpolating polynomial

$$P(x) = f(x_0) \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + f(x_1) \frac{(x-x_0)(x-x_2)}{(x_1-x_2)(x_1-x_0)} + f(x_2) \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

Prove that

$$\int_{x_0}^{x_2} f(x) dx = \int_{x_0}^{x_2} P(x) dx$$

Composite integration methods

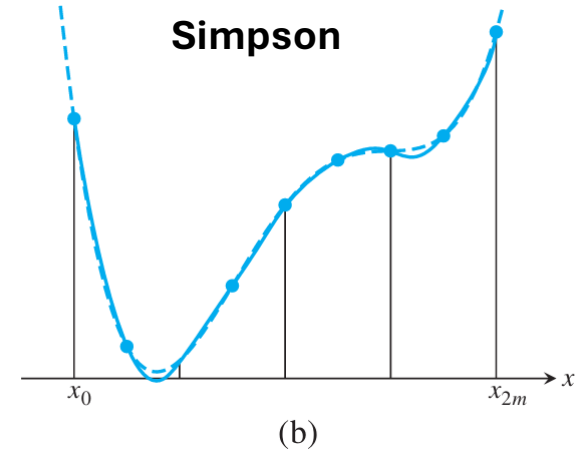
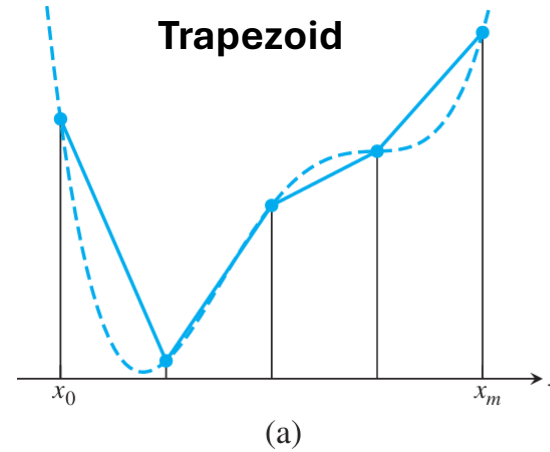
(Integration using multiple panels)

Trapezoidal method

$$\int_{x_i}^{x_{i+1}} f(x) dx = \frac{h}{2} (f(x_i) + f(x_{i+1}))$$

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(a) + f(b) + 2 \sum_{i=1}^{m-1} f(x_i) \right]$$

$$\text{Error} = - \sum_{i=0}^{m-1} \frac{h^3}{12} f''(c_i) = - \frac{(b-a)h^2}{12} f''(c)$$



Simpson's method

$$\int_{x_{2i}}^{x_{2i+2}} f(x) dx = \frac{h}{3} [f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})]$$

$$\int_a^b f(x) dx = \frac{h}{3} \left[f(a) + f(b) + 4 \sum_{i=1}^m f(x_{2i-1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) \right]$$

$$\text{Error} = - \frac{(b-a)h^4}{180} f^{(iv)}(c),$$